

Stream Forelle — Sensor Hardware Budget

FunkyForelle · CHF 12,000

Stream Forelle is responsible for sensor selection, procurement, calibration, and deployment across 4 monitoring sites on Swiss rivers: 1 full multi-parameter pilot station and 3 temperature + depth loggers. Based at OST Rapperswil (Lake Zurich campus). University lab resources used at no cost.

Item	Qty	Unit CHF	Total CHF
PILOT SITE — Patrick's river (full multi-parameter)			
In-Situ Aqua TROLL 500 (DO + temp + conductivity + depth)	1	3,500	3,500
Telemetry module (GSM) + weatherproof housing	1	800	800
Solar panel + battery backup + mounting hardware	1	800	800
ADDITIONAL SITES — temperature + depth only			
Keller Series 36W (temp + depth logger)	3	500	1,500
Basic telemetry module	3	300	900
SHARED COSTS			
Field installation — travel, permits, 4 sites	1	1,500	1,500
SIM cards + data plans (4 sites × 4 months)	1	400	400
Consumables — cables, connectors, O-rings	1	300	300
RESERVE			
Sensor failure / replacement contingency	1	2,300	2,300
TOTAL			12,000
University resources at no cost (OST lab, 3D printer, calibration equipment)			(0)
Budget envelope			12,000

Notes

- OST Rapperswil IoT lab provides calibration equipment, 3D printing, and IoT components at no cost to enrolled students.
- Patrick Hager (Fischereiverein See + Gaster) advises on site selection at no charge.
- Simplified design: 1 full multi-parameter pilot site + 3 temp/depth validation sites.
- All hardware documented and open-sourced as project deliverables.
- Prices indicative (September 2026); final selection subject to intern's market analysis.